Detailed search algorithm
Roesch et al. – Mitochondrial disease and hearing loss in children – a systematic review

Databases searched:

- Ovid Medline (MEDLINE); 24 segments; 1946 – September 2021
- Ovid Cochrane Central Register of Controlled Trials (CCTR (formerly Cochrane Controlled Trials Register)); 1991 – September 2021
- Ovid Embase (Excerpta Medica Database); 1974 – September 2021

Applied search strategies:

a) Medline:

1 (mitochondrial disease? or mitochondriopath?).mp.
2 (mitochondrial adj15 disease).mp.
3 exp Mitochondrial Diseases/
4 1 or 2 or 3
5 hearing loss.mp.
6 (hearing adj15 loss).mp.
7 exp Hearing Loss/
8 5 or 6 or 7
9 (audiometry or audiometric testing).mp.
10 exp audiometry
11 9 or 10
12 8 or 11
13 4 and 12
14 limit 13 to ("all child (0 to 18 years)" and (english or german))

b) CCTR:

1 (mitochondrial disease? or mitochondriopath?).mp. [mp=title, original title, abstract, mesh headings, heading words, keyword]
2 (mitochondrial adj15 disease).mp. [mp=title, original title, abstract, mesh headings, heading words, keyword]
3 exp Mitochondrial Diseases/
4 1 or 2 or 3
5 hearing loss.mp. [mp=title, original title, abstract, mesh headings, heading words, keyword]
6 (hearing adj15 loss).mp. [mp=title, original title, abstract, mesh headings, heading words, keyword]
7 exp Hearing Loss/
8 5 or 6 or 7
9 (audiometry or audiometric testing).mp. [mp=title, original title, abstract, mesh headings, heading words, keyword]
10 exp Audiometry, Evoked Response/ or exp Audiometry/ or exp Audiometry, Speech/ or exp Audiometry, Pure-Tone/
11 9 or 10
12 8 or 11
13 4 and 12

c) Embase:

1 (mitochondrial disease? or mitochondriopath?).mp.
2 (mitochondrial adj15 disease).mp.
3 exp "disorders of mitochondrial functions"/
4 1 or 2 or 3
5 hearing loss.mp.
6 (hearing adj15 loss).mp.
7 exp hearing impairment/
8 5 or 6 or 7
9 (audiometry or audiometric testing).mp.
10 exp audiometry/
11 9 or 10
12 8 or 11
13 4 and 12
14 limit 13 to ((english or german) and (infant <to one year> or child <unspecified age> or preschool child <1 to 6 years> or school child <7 to 12 years> or adolescent <13 to 17 years>))

d) Additional hand searching of reference lists of articles chosen after primary, computer-based search

Concerning CCTR database (b), limits such as patients under the age of 18 years and languages German and English cannot be applied.